

BYTEBANDITS • AI TRACK

Container Congestion Predictor & Port Operations Optimiser

From real vessel-call data to proactive, shift-ready port decisions

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Port congestion is discovered too late

WHY IT HURTS OPERATORS

- Planning still runs on manual spreadsheets
- Vessel queues build before capacity problems are visible
- Berth and crane coordination is hard to align
- Supervisors need immediate, actionable decisions

WHEN VISIBILITY COMES TOO LATE

2021 LA/LB backlog: 100+ ships held offshore for weeks

Vessel arrivals

Manual planning

Late congestion discovery

Delays and waiting

THE SOLUTION

One operational view connects arrivals, capacity, and action

- 01** IMF PortWatch AIS-based vessel-call data
- 02** 72-hour calibrated vessel simulation
- 03** Congestion hotspot detection
- 04** Greedy berth & crane assignment
- 05** AI-generated operations brief
- 06** React operations dashboard

Operations dashboard — simulated per-vessel details labeled



Real aggregate data in, explainable decisions out

DATA

IMF PortWatch API → daily vessel-call aggregates → CSV cache / offline fallback

BACKEND

FastAPI → simulator.py → congestion.py → optimizer.py → brief_generator.py

OUTPUTS

72-hour congestion timeline · hotspot severity summary · berth & crane assignments · vessel wait times · Gemini or rule-based operations brief

FRONTEND

React + Vite + Recharts dashboard

DEPLOYMENT

Vercel frontend → Render backend

DEMO / KEY FEATURE

A shift supervisor can see the next 72 hours at a glance

Berth & Crane Assignment Schedule

55 vessels

All (55)

Assigned (11)

Reschedule (44)

Reroute (0)

55 vessels shown

ID	VESSEL	LINE	CLASS	TEU	ORIGIN	ETA	BERTH	CRANES	START	END	WAIT	STATUS
VB42	YAN390	Yang Ming	Ultra-Large (ULCS)	18,000	Tokyo	15 Sept, 11:13	Busan New Port HPNT	4	15 Sept, 11:13	17 Sept, 01:13	0.0h	Assigned
VB16	COS949	COSCO	Feeder	2,000	Tokyo	15 Sept, 11:44	Busan New Port BNCT	4	15 Sept, 11:44	15 Sept, 22:44	0.0h	Assigned
VB39	CMA654	CMA CGM	Post-Panamax	10,000	Rotterdam	15 Sept, 12:10	Busan New Port PNIT	3	15 Sept, 12:10	16 Sept, 11:10	0.0h	Assigned
VB49	HAP498	Hapag-Lloyd	Feeder	2,000	Ningbo	15 Sept, 12:12	Busan Newport BPA T4	3	15 Sept, 12:12	16 Sept, 00:12	0.0h	Assigned
VB55	HMM640	HMM	Feeder	2,000	Tokyo	15 Sept, 16:52	Gamcheon Container T	2	15 Sept, 16:52	16 Sept, 05:52	0.0h	Assigned
VB33	COS590	COSCO	Feeder	2,000	Rotterdam	15 Sept, 19:59	Uam Container T	2	15 Sept, 19:59	16 Sept, 05:59	0.0h	Assigned
VB01	YAN555	Yang Ming	New-Panamax	14,000	Tianjin	15 Sept, 11:50	Busan New Port BNCT	4	15 Sept, 22:44	17 Sept, 05:44	10.9h	Assigned
VB45	HMM680	HMM	Post-Panamax	10,000	Qingdao	15 Sept, 12:28	Busan Newport BPA T4	3	16 Sept, 00:12	16 Sept, 20:12	11.7h	Assigned
VB05	HAP718	Hapag-Lloyd	Feeder	2,000	Los Angeles	15 Sept, 22:04	Gamcheon Container T	2	16 Sept, 05:52	16 Sept, 16:52	7.8h	Assigned
VB52	MSC915	MSC	Feeder	2,000	Tokyo	15 Sept, 22:40	Uam Container T	2	16 Sept, 05:59	16 Sept, 12:59	7.3h	Assigned
VB08	EVE289	Evergreen	Panamax	5,000	Tianjin	15 Sept, 12:54	Busan New Port PNIT	3	16 Sept, 11:10	17 Sept, 05:10	22.3h	Reschedule
VB36	HAP636	Hapag-Lloyd	Post-Panamax	10,000	Shanghai	15 Sept, 14:08	Busan Newport BPA T4	3	16 Sept, 20:12	17 Sept, 20:12	30.1h	Reschedule
VB37	YAN200	Yang Ming	Ultra-Large (ULCS)	18,000	Tokyo	15 Sept, 11:20	Busan New Port HPNT	4	17 Sept, 01:13	18 Sept, 15:13	37.9h	Reschedule
VB14	ONE113	ONE	Post-Panamax	10,000	Rotterdam	15 Sept, 14:40	Busan New Port PNIT	3	17 Sept, 05:10	18 Sept, 01:10	38.5h	Reschedule
VB48	COS987	COSCO	New-Panamax	14,000	Los Angeles	15 Sept, 14:39	Busan New Port BNCT	4	17 Sept, 05:44	18 Sept, 11:44	39.1h	Reschedule
VB04	MSC183	MSC	Panamax	5,000	Los Angeles	15 Sept, 15:53	Busan Newport BPA T4	3	17 Sept, 20:12	18 Sept, 12:12	52.3h	Reschedule
VB29	EVE301	Evergreen	Panamax	5,000	Rotterdam	15 Sept, 17:05	Busan New Port PNIT	3	18 Sept, 01:10	18 Sept, 20:10	56.1h	Reschedule

Congestion timeline — demand, capacity, severity

Assignment table — berth, crane count, start/end, wait

Operations brief — concise, shift-ready plan

Detect the hotspot before the queue becomes the problem.

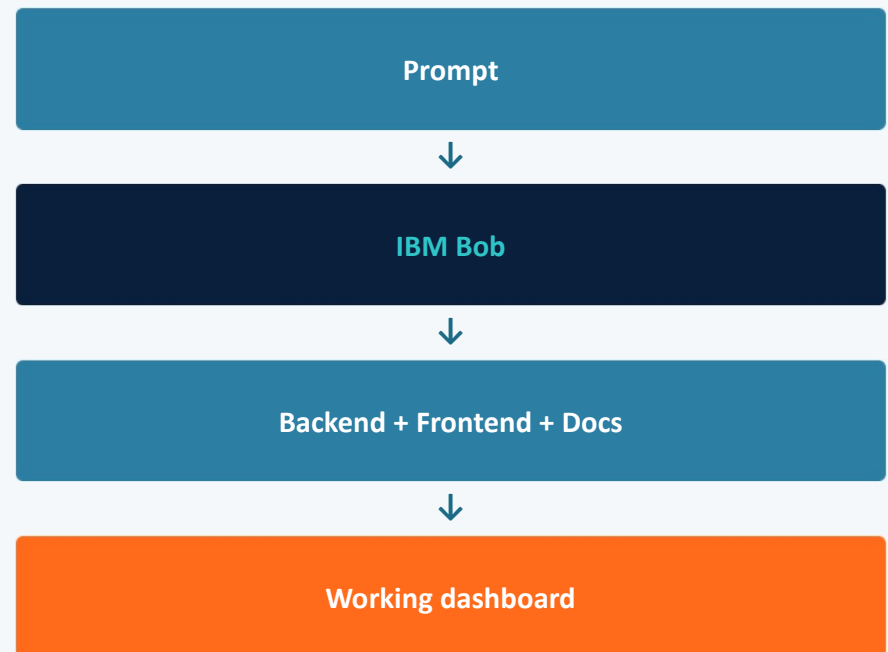
IBM Bob was load-bearing throughout the build

HOW WE USED IBM BOB

- AI coding agent and development platform used end-to-end
- Built and iterated the Python backend inside Bob
- Built the React frontend inside Bob
- Refined configuration and documentation with Bob

TRANSPARENCY

Google Gemini powers the optional natural-language operations brief and is not IBM technology. IBM Bob is our AI development platform and coding agent.



RESULTS / IMPACT

The result is earlier visibility and explainable action

DEFENSIBLE RESULTS

- 72-hour forward planning window
- One-hour congestion monitoring slots
- LOW / MEDIUM / HIGH hotspot classification
- Greedy berth assignment, $O(V \times B)$ complexity
- ~100 vessels handled in under 50 ms (architecture notes)
- Structured operations brief for shift supervisors
- Offline fallback when live data or Gemini is unavailable

CURRENT LIMITATIONS

- Per-vessel ETAs and berth/crane counts are simulated
- Greedy FCFS optimisation is not globally optimal
- Gemini requires an API key
- Render free-tier cold starts take 30–60s

THE TEAM

ByteBandits

Srujal Patel

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[Role to confirm]

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Turning vessel data into decisions before congestion becomes delay.

